

CARPENTERS FIJI PTE LIMITED

MID-TERM REVIEW REPORT

Car Rental and Equipment Management System (CREMS)

Document purpose: This mid-term review reports on the progress of the CREMS project, including the requirements baseline, system architecture, project plan updates, risk assessment, and timeline for remaining deliverables.

Document control	Value
Client	Carpenters Fiji Pte Limited
Project	Car Rental and Equipment Management System (CREMS)
Course	CS400 Industry Experience Project - S2 2026
Group number	10
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Version	1.0 - Mid-Term Review
Document date	1 September 2026
Classification	Client confidential - project use

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1. Introduction

This mid-term review reports on the progress of the Car Rental and Equipment Management System (CREMS) project being developed for Carpenters Fiji Pte Limited as part of the CS400 Industry Experience Project at the University of the South Pacific.

The review covers work completed during Weeks 1-5 of the 14-week project timeline (3 August - 1 September 2026). It presents the approved requirements baseline, system architecture, project plan updates, milestones achieved, obstacles encountered, risk mitigation strategies, and the timeline for completing remaining deliverables.

The CREMS project aims to deliver a configurable, group-wide platform for managing rental revenue, asset maintenance expenditure, and customer service across Carpenters Fiji divisions, with initial focus on Carpenters Motors and Carptrac operations.

2. Requirements Document (SRS)

The Software Requirements Specification (SRS) document was completed and approved as Version 1.0 on 18 August 2026. It serves as the controlling baseline for design, development, testing, user acceptance, and handover throughout the project.

2.1 SRS Overview

The SRS defines the complete business, functional, data, security, and quality requirements for the CREMS Motors and Carptrac MVP. Key sections include:

- Introduction and product vision for the CREMS platform
- Stakeholders and user classes (Public visitor, Customer, Administrator, Branch Manager, Rental Officer, Maintenance Officer, Finance Officer, Driver/Operator)
- Scope baseline confirming in-scope and out-of-scope capabilities
- 130+ functional requirements across 10 domains (Identity, Organization, Customers, Assets, Bookings, Rentals, Inspections, Maintenance, Personnel, Finance)
- 15 core business rules governing rental operations
- Data requirements covering all operational domains
- External interfaces (Web UI, REST API, Database, Email, QR scanner, PDF generation)
- Non-functional requirements (Security, Performance, Reliability, Usability, Maintainability)
- Reporting requirements and acceptance criteria
- 12 open decisions requiring Carpenters confirmation

2.2 Requirements Summary

Requirement Domain	Total	Must	Should
Identity and Access (IAM)	15	13	2
Organization and Divisions (ORG)	10	8	2
Customer Management (CUS)	10	7	3
Asset Register (AST)	15	9	6
Quotation and Booking (BKG)	13	10	3
Rental Operations (HIR)	11	11	0
Inspections (INS)	11	10	1
Maintenance (MNT)	10	5	5
Personnel and Field (OPS)	6	2	4
Finance and Reporting (FIN)	14	9	5

Full SRS Document: The complete Software Requirements Specification v1.0 is included as Appendix A of this report and is available as a standalone document:
[CREMS_Software_Requirements_Specification_v1.0.pdf](#)

3. System Architecture and Design

The Functional Specification Document (FSD) translates the approved requirements into observable system behavior and defines the technical architecture for implementation.

3.1 Architectural Overview

CREMS uses a modular monolith architecture: one React frontend, one ASP.NET Core API, and one SQL Server database. This approach keeps deployment and debugging manageable while retaining clean API boundaries for future extensions.

Layer	Technology	Responsibility
Frontend	React 19, TypeScript, Vite, Material UI	Responsive public site, customer portal, task-based staff portal
API	ASP.NET Core 10 Web API	Validation, authorization, workflow orchestration, REST contracts
Identity	ASP.NET Core Identity, secure cookies	Staff/customer separation, roles, recovery, session security
Data	SQL Server 2022, Entity Framework Core	Relational integrity, transactions, migrations, reporting
Delivery	Docker Compose, GitHub, Postman	Repeatable environments, collaboration, testing

3.2 Backend Modules

The system is organized into the following domain modules:

- Identity and access management
- Branches and organization
- Customers and portal management
- Assets and catalogue
- Bookings and availability
- Rentals and returns
- Inspections and damage
- Maintenance and costs
- Reporting and dashboards
- Notifications and email
- Auditing and history

3.3 Key Design Decisions

Decision	Rationale
GUID primary keys	Avoid exposing predictable record counts; simplify offline imports
DateTimeOffset for events	Proper time zone handling; Fiji-local conversion at boundaries
Explicit asset states	Clear operational state rather than deriving from UI labels

Decision	Rationale
Booking lines carry agreed rate	Master-rate changes do not alter historical records
SQL constraints + transactional checks	Joint prevention of double booking
Soft delete via deactivation	Preserve audit history and referential integrity

Full FSD Document: The complete Functional Specification Document v1.0 is included as Appendix B of this report and is available as a standalone document:
CREMS_Functional_Specification_Document_v1.0.pdf

4. Project Plan Updates

The project is planned as a 12-week agile delivery from 10 August to 30 October 2026, organized into one-week sprints with weekly client checkpoints. The following sections detail concrete steps taken during the first four weeks.

4.1 Work Completed (Weeks 1-4)

Week 1: Scope and Foundation (10-14 August)

- Client meeting and domain research
- Approved Motors/Carptrac boundary confirmation
- Requirements baseline and risk register establishment
- Git repository and branch conventions setup
- Solution skeleton and initial SQL Server container configuration

Week 2: Authentication and Startup Services (17-21 August)

- Staff/customer sign-in implementation
- Secure password handling with ASP.NET Identity
- Cookie/session timeout configuration
- API authorization defaults
- Health checks and error handling foundations

Week 3: Organization and Access Isolation (24-28 August)

- Division/branch administration
- User assignments and role matrix
- API-level division and branch isolation
- Configuration screens for organization setup

Week 4: Customer Module (31 August - 4 September)

- Individual and corporate customer profiles
- Contacts and approved renters management
- Customer portal foundation
- Validation and search functionality

5. Milestones Achieved

Milestone	Sprint	Target Date	Status
M1 — Scope and engineering foundation	Week 1	14 August 2026	Completed
M2 — Secure sign-in demo	Week 2	21 August 2026	Completed
M3 — Access-isolation demo	Week 3	28 August 2026	Completed
M4 — Configurable asset-register demo	Week 5	11 September 2026	In Progress
M5 — Quotation-to-booking demo	Week 7	25 September 2026	Pending
M6 — Motors lifecycle demo	Week 8	2 October 2026	Pending
M7 — Carptrac lifecycle demo	Week 10	16 October 2026	Pending
M8 — Final handover	Week 12	30 October 2026	Pending

Milestone Status: The project is on track with the planned schedule. Milestones 1-3 were completed on time. Milestone 4 is currently in progress and on schedule for completion by 11 September 2026.

6. Obstacles Encountered and Mitigation

6.1 Requirements Uncertainty

Several business rules remain subject to client confirmation, including pricing rules, approval thresholds, and inspection requirements.

Mitigation:

- Maintained a requirements register tracking 18 open decisions
- Documented provisional assumptions where implementation must begin early
- Scheduled fortnightly client checkpoints for decision confirmation
- Designed configurable rules to accommodate various pricing and approval models

6.2 Scope Management

The group-wide modular scope presents complexity in designing a system that supports multiple divisions without duplicating code or data.

Mitigation:

- Adopted capability-driven division configuration
- Implemented progressive disclosure for customer journeys
- Protected MVP boundary with strict change control
- Used feature flags for division-specific capabilities

6.3 Technical Complexity

The dual customer/staff authentication model with division/branch scoping adds complexity to authorization and data access.

Mitigation:

- Implemented CurrentStaffScope service for consistent scope enforcement
- Created granular permission system beyond basic roles
- Used server-side query filtering to prevent data leakage
- Developed comprehensive authorization test suite

6.4 Client Availability

Scheduling client meetings and obtaining timely decisions on business rules has been challenging.

Mitigation:

- Prepared focused decision requests with clear options and defaults
- Used requirements register to prioritize decisions by impact
- Documented provisional assumptions to maintain development momentum
- Scheduled regular fortnightly checkpoints

7. Changes to the Plan

The following changes have been made to the original project plan since its approval:

7.1 Scope Clarification

Following the first client meeting, the scope was clarified to emphasize group-wide modular capabilities rather than a vehicle-only rental system. This led to:

- Addition of configurable division capabilities and service offerings
- Implementation of personnel requirement configuration for equipment hire
- Enhanced customer model supporting both individual and corporate accounts
- Progressive disclosure design for public customer journeys

7.2 Architecture Refinement

The architecture was refined based on implementation experience:

- Adopted window-bound session security model
- Implemented staged email delivery with redirect-all for testing
- Added comprehensive permission system beyond basic roles
- Configured division-aware staff scope with branch assignment

7.3 Schedule Adjustments

No significant schedule adjustments have been required. The project remains on track with the original timeline. Minor adjustments include:

- Overlapping work packages to maintain continuous progress
- Early start on asset/customer modules to validate architecture decisions

8. Risk Management Update

8.1 Current Risk Register

ID	Risk	P	I	Score	Status	Mitigation
R1	Requirements unclear/change late	4	5	20 H	Active	Decision log, change control, fortnightly validation
R2	Scope exceeds capacity	4	5	20 H	Active	MVP protection, MoSCoW priorities
R3	Authorization data leakage	3	5	15 H	Mitigated	Server scope queries, negative tests, peer review
R4	Booking rules incorrect	3	5	15 H	Active	Transactions, overlap tests, domain rules
R5	Equipment safety incomplete	3	5	15 H	Active	Client validation, mandatory gates
R6	Credential exposure	2	5	10 M	Mitigated	Synthetic data, user secrets, scanning
R7	Member unavailability	3	4	12 M	Active	Pairing, shared docs, cross-training
R8	Late integration	3	4	12 M	Active	CI, fortnightly demos
R9	Client availability delays	3	4	12 M	Active	Early booking, focused questions
R10	Data loss	2	5	10 M	Mitigated	GitHub remote, backup rehearsal
R11	Email misconfiguration	2	4	8 M	Active	Staging redirect-all, delivery log
R12	UI complexity	3	4	12 M	Active	Task language, progressive disclosure
R13	Report reconciliation	3	4	12 M	Active	Defined formulas, reconciliation tests
R14	Deployment owner unknown	3	4	12 M	Active	Decision by W11, portable staging
R15	Schedule delay	3	4	12 M	New	Scope substitution, 80% iteration target

8.2 New Risks Identified

One new risk has been identified since the original plan:

ID	Risk	P	I	Score	Mitigation
R15	Schedule delay due to requirements uncertainty	3	4	12 M	Scope substitution policy, 80% iteration target, escalation after 2-day blockers

8.3 Risk Mitigation Progress

The following risks have been effectively mitigated through implementation:

- R3 (Authorization): Implemented comprehensive server-side scope enforcement and negative testing
- R6 (Credentials): Established secret management, synthetic data policies, and repository scanning
- R10 (Data loss): Configured GitHub remote, protected branches, and backup/restore procedures

9. Timeline for Remaining Deliverables

Week	Dates	Focus	Exit Deliverable
W5	7-11 Sep	Complete asset module, categories, dynamic attributes	M4: Asset-register demo
W6	14-18 Sep	Availability, rates and quotations	Rate configuration demo
W7	21-25 Sep	Bookings, approvals, agreements, notifications	M5: Quotation-to-booking demo
W8	28 Sep-2 Oct	Pre-hire, QR check-out, return, post-hire inspection	M6: Motors lifecycle demo
W9	5-9 Oct	Maintenance, damage and detailed expenditure	Maintenance module demo
W10	12-16 Oct	Carptrac personnel and equipment-hire variations	M7: Carptrac lifecycle demo
W11	19-23 Oct	Dashboards, profitability, integration and UAT	Feature-complete build
W12	26-30 Oct	Stabilization, handover and closure	M8: Final handover

9.1 Critical Path

The critical path for remaining work includes:

- Asset/service catalogue completion (W6)
- Availability and booking rules (W7-8)
- Agreement and checkout workflow (W9)
- Return and maintenance processing (W10)
- Client UAT and acceptance (W13)

9.2 Upcoming Client Decisions

The following client decisions are required to maintain schedule:

Decision	Required By	Impact if Delayed
Pricing and approval rules	Before W7	Cannot implement quotation engine
Inspection template confirmation	Before W9	Cannot finalize checkout workflow
Corporate account rules	Before W7	Cannot complete corporate features
Deployment infrastructure	Before W11	Cannot prepare production environment

10. Conclusion

The CREMS project has completed its first five weeks on schedule with all planned milestones achieved. The requirements baseline has been established, the system architecture is implemented, and significant progress has been made on identity, access management, and core domain entities.

Key achievements include:

- Approved SRS document with 130+ functional requirements across 10 domains
- Implemented secure authentication with role-based access and division/branch scoping
- Established modular architecture supporting configurable division capabilities
- Developed comprehensive FSD translating requirements into observable behavior
- Maintained effective client communication through fortnightly checkpoints

The primary risks remain requirements uncertainty and scope management, both of which are being actively mitigated through the requirements register, change control, and client engagement processes. No schedule adjustments are currently required.

The team remains confident in delivering the CREMS MVP within the planned 14-week timeline, meeting all academic and client acceptance criteria.

Appendix A - SRS Summary

The complete Software Requirements Specification v1.0 (CREMS_Software_Requirements_Specification_v1.0.pdf) is available alongside this report. Key sections include:

Section	Description
1. Introduction	Purpose, product vision, definitions, and requirement priority language
2. Stakeholders	User classes and their primary responsibilities
3. Scope Baseline	In-scope, out-of-scope, assumptions, and dependencies
4. Functional Requirements	130+ requirements across 10 functional domains
5. Business Rules	15 core business rules governing operations
6. Data Requirements	Logical data structure and client-supplied data mapping
7. External Interfaces	Web UI, REST API, database, email, QR scanner, PDF
8. Non-Functional Reqs	Security, performance, reliability, usability, maintainability
9. Reporting	Role-specific dashboards and reports
10. Acceptance	UAT scenarios and traceability
11. Open Decisions	12 decisions requiring Carpenters confirmation
12. Change Control	Change management process
13. Client Sign-off	Approval and signature section

Appendix B - Design Document Summary

The complete Functional Specification Document v1.0 (CREMS_Functional_Specification_Document_v1.0.pdf) is available alongside this report. Key sections include:

Section	Description
1. Introduction	Purpose, in-scope divisions, functional boundaries
2. Architecture	System context, functional design principles
3. Roles & Permissions	Role definitions and permission rules
4. UI Behavior	Common user interface patterns and standards
5. Authentication	Staff/customer auth, sessions, access administration
6. Organization	Divisions, branches, services configuration
7. Customers	Customer records and portal functions
8. Assets	Asset register, QR, lifecycle, availability
9. Bookings	Requests, quotations, approvals workflow
10. Rentals	Pickup, agreements, active hire, returns
11. Maintenance	Inspections and maintenance processing
12. Personnel	Operators, drivers, field operations
13. Finance	Invoicing, profitability, dashboards
14. Notifications	Email, documents, scheduled processing
15. API/Validation	API conventions, errors, audit behavior
16. Acceptance	Functional acceptance scenarios
17. Assumptions	Assumptions, decisions, change control
18. Sign-off	Client approval section